Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-C-CLASS-P0113
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz C-Class (W206 / C 300)
Powertrain: 2.0L Turbocharged Inline-4 (M254) with 2nd Gen ISG (Beltless 48V Mild-Hybrid)
Engine Control System: Bosch MD1CS006 (Engine Control Module N3/10) - High-Speed CAN-E2 Architecture
Electrical Architecture: Integrated 12V / 48V Architecture with 48V DC/DC Converter (N83/1)
Fuel Injection Architecture: Single Cam-Driven High Pressure Pump (HPP) with Solenoid Direct Injectors (200-250 bar)
Firing Order: 1-3-4-2

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0113`
- **Standard Fault Description:** Intake Air Temperature (IAT) Sensor 1 Circuit High Input
- **System Definition:** IAT thermistor circuit voltage exceeds 4.85V, indicating an open circuit or unplugged harness connector.
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz C-Class (W206 / C 300)**:

1. **Charge Air Temperature Sensor B17/2 Ground Wire Chafed:** Harness harness clipped too tightly near alternator bracket, wire severed inside insulation.
2. **Sensor Thermistor Pellet Burnout:** Extreme heat cycle failure after sustained high boost driving.
3. **Front SAM Module Fuse F14 Micro-crack:** Intermittent power/reference dropout causing ADC high saturation.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the C-Class control harness:

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[Charge Air Temp Sensor B17/2]             [ECM N3/10 Connector 'K']
Pin 1 (Sensor Ground Reference) ---------> Pin 78 (GND_SENSOR)
Pin 2 (Thermistor Signal / 5V Pull-up) --> Pin 79 (ADC Temp Signal)
```

- Voltage Characteristic: Pin 2 to Pin 1: 4.88V to 5.00V indicates open circuit (default fail-safe reading = -40°C). Normal ambient 25°C = ~2.2V.

Electrical Reference Points

- **Primary Ground Reference:** `W10 (Engine Block Ground Point, Right) & W9 (Front Wheel Arch Ground)`
- **Power Distribution Node:** `Engine Bay Prefuse Box F32/1 & Front SAM Module (N10/1)`
- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

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4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Check freeze-frame parameter: Intake air temperature reading. If locked at -40°C, circuit is fully open.
2. Backprobe Pin 2 with digital multimeter; verify presence of 5.0V pull-up from ECM N3/10.
3. Check Pin 1 continuity to ECM Pin 78 (< 0.5Ω).
4. Inspect harness routing across the front radiator support bracket for pinch points.

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5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace Charge Air Temperature Sensor B17/2 (twist-lock sensor on charge pipe).
- Re-route and sleeve wiring harness with convoluted tubing and friction tape away from sharp bracket edges.
- Clear DTC and road test to verify IAT tracks ambient + 10-25°C under normal boost.

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6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.